Pointer arithmetic

Pointer arithmetic means changing the value of a pointer to make it point to a different element in memory.

```
#include <stdio.h>

int main() {
  int myNumbers[4] = {25, 50, 75, 100};
  int *p = myNumbers;  // points to address of myNumbers[0]

  printf("%d\n", *p);     // 25
  printf("%d\n", *(p + 1)); // 50
  printf("%d\n", *(p + 2)); // 75
  printf("%d\n", *(p + 3)); // 100
  return 0;
}
25
50
75
100
```

```
#include <stdio.h>

int main() {
  int myNumbers[4] = {25, 50, 75, 100};
  int *p = myNumbers;  // points to myNumbers[0]
  printf("%p\n", p);
  printf("%p\n", p+1);
  printf("%p\n", p+2);
  printf("%p\n", p+3);
  return 0;
}
0xffffd00daa18
0xffffd00daa1c
0xffffd00daa20
0xffffd00daa24
```

```
#include <stdio.h>

int main(){
    int arr[] = { 1, 2, 3, 4, 5 };
    int* ptr=arr;
    for (int i = 0; i < 5; i++) {
        printf("%p\n", &ptr[0]);
        ptr++;
    }
}
0xffffdefc94a0
0xffffdefc94a4
0xffffdefc94a8
0xffffdefc94ac
0xffffdefc94b0
```

```
#include <stdio.h>
```

```
int main(){
    int arr[] = { 1, 2, 3, 4, 5 };
    int* ptr=arr;
    for (int i = 0; i < 5; i++) {
        printf("%d ", ptr[0]);
        ptr++;
    }
}
```
1 2 3 4 5

Pointer Comparison

```
#include <stdio.h>

int main(){
    int arr[5];
    int* ptr1 = arr;
    int* ptr2 = &arr[0];
    if (ptr1 == ptr2) {
        printf("Pointer to Array Name and First Element are Equal.");
    }
    else {
        printf("Pointer to Array Name and First Element are not Equal.");
    }

    return 0;
}
```
Pointer to Array Name and First Element are Equal.

Pointer Increment and Decrement

You can move a pointer with ++ and -- (and with += / -=):

```
#include <stdio.h>

int main() {
  int myNumbers[3] = {10, 20, 30};
  int *p = myNumbers;

  printf("%d\n", *p);
  p++;
  printf("%d\n", *p);
  p--;
  printf("%d\n", *p);

  p += 2;        // Pointer Addition
  printf("%d\n", *p);

  p -= 2;        // Pointer subtraction
  printf("%d\n", *p);
  return 0;
}
```
10

```
20
10
30
10
```

Pointer Subtraction

```
#include <stdio.h>

int main(void) {
    int arr[] = {10, 20, 30, 40, 50};
    int *ptr = &arr[3];

    printf("Before subtraction: %p\n", ptr);

    ptr = ptr - 2;

    printf("After subtraction:  %p\n", ptr);
    printf("Value: %d\n", *ptr);

    return 0;
}
```

```
Before subtraction: 0xffffd038ec5c
After subtraction:  0xffffd038ec54
Value: 20
```

Pointer Subtraction (Distance)

You can subtract two pointers that point to elements in the **same array** to find out how many elements are between them:

```
#include <stdio.h>

int main() {
  int myNumbers[5] = {10, 20, 30, 40, 50};
  int *start = &myNumbers[1];
  int *end = &myNumbers[4];

  printf("%td\n", end - start); // 3 elements apart
  return 0;
}
```

```
3
```

The following operations are not allowed using pointers

1. Multiplication p * 2 Not allowed
2. Division p / 2 Not allowed
3. Modulo p % 2 Not allowed
4. Adding two pointers p + q Not allowed
5. Multiplying pointers p * q Not allowed

6. Dividing pointers p / q Not allowed

7. Modulo with pointers p % q Not allowed

Pointer arithmetic in C supports:

- Pointer + integer
- Pointer − integer
- Pointer increment and decrement
- Subtraction of two compatible pointers into the same array
- Pointer comparisons, subject to the rules of C

Pointer arithmetic does not support:

- Pointer + pointer
- Pointer × integer
- Pointer ÷ integer
- Pointer % integer
- Multiplication, division or modulo between pointers